

PhD Application Guide — One-Page Summary

Yigit Onat · May 2026 · Preprint: *Event-Aligned SVD for Pattern Kernel Discovery in Crypto Perpetual Futures*

Publishing Platforms

SSRN (ssrn.com) — No affiliation required. Free. Upload PDF, get permanent link in 1–3 days. Standard venue for finance preprints. Use *Capital Markets: Market Microstructure* e-journal.

arXiv (arxiv.org) — No affiliation required but needs endorsement from existing author (spam filter, not peer review). Categories: `q-fin.TR`, `q-fin.ST`, `stat.ML`.

Strategy: SSRN first (no barriers) → email professors with link → get arXiv endorsement → cross-post.

ETH Zürich

Josef Teichmann – D-MATH. ML in finance, deep hedging, rough analysis. *Best overall fit*. Pitch: latent structure discovery in LOB dynamics via SVD.

Beatrice Acciaio – D-MATH, SFI. Optimal transport, robust finance. Pitch: Wasserstein regime detection on order book distributions.

Patrick Cheridito – D-MATH. Deep learning for optimal stopping, approximation theory. Pitch: approximation bounds on SVD-extracted features.

Peter Bühlmann – D-MATH Statistics. High-dim inference, stability selection, FDR. Pitch: replicability guarantees for microstructure signals.

Helmut Bölcskei – D-ITET. Signal processing, harmonic analysis, ML theory. *Best methods fit*. Pitch: info-theoretic limits of pattern recovery from noisy LOB data.

Walter Farkas – D-MATH + UZH. Director MSc Quant Finance. Crypto theses supervised. Pitch: empirical microstructure of crypto perpetuals.

Markus Leippold – UZH/SFI. NLP in finance, crypto markets. Most direct crypto expertise in Zurich. Co-supervision with ETH feasible.

EPFL Lausanne

PhD via EDFI program (SFI). Deadlines: **Jan 15** and **Mar 31**.

Pierre Collin-Dufresne – SFI. Microstructure, informed trading, order flow. *Best microstructure fit*. Ex-Goldman. Pitch: SVD detection of informed trading patterns.

Julien Hugonnier – SFI. Derivatives, OTC microstructure. Published “Perpetual Futures Pricing” (Math Finance 2025). Pitch: empirical microstructure grounded in perp pricing theory.

Semyon Malamud – SFI. ML + asset pricing, “AI Pricing Theory” (w/ Bryan Kelly). Pitch: SVD factors as inputs to learned pricing kernels.

Damir Filipovic – Swissquote Chair. Stochastic processes, ML in quant finance. Ex-Princeton, PhD ETH. Pitch: regime-switching models with SVD state variables.

Lenka Zdeborova – SPOC Lab. Statistical physics of ML, phase transitions, signal recovery. Pitch: phase transitions in microstructure signal recovery. Needs SFI co-advisor.

Florent Krzakala – IdePHICS Lab. Random matrix theory, spiked models, compressed sensing. Pitch: spiked matrix models for LOB feature matrices. Needs SFI co-advisor.

Daniel Kuhn – RAO Lab. Distributionally robust optimization. 2024 Farkas Prize. Pitch: robust execution/sizing for pattern-based strategies.

Top Pairings

Finance + Methods:	Teichmann + Bölcskei (ETH)
Microstructure + RMT:	Collin-Dufresne + Krzakala (EPFL)
ML pricing + Physics:	Malamud + Zdeborova (EPFL)

Email Template (Key Lines)

Subject: Prospective PhD — Microstructure Pattern Discovery via SVD

Mention: (1) embedded systems + quant trading background, (2) preprint on SSRN with link, (3) key result: 6 significant kernels from 158 candidates after BH-FDR, turtle-soup/trap IC $\in [0.20, 0.36]$, (4) bridge to professor’s specific work, (5) offer to share codebase.